

Revision Notes for Class 7 Maths Chapter 2 – Operations with Integers

Integers are whole numbers that can be positive, negative, or zero. In Class 7 Maths Chapter 2, you review and extend your knowledge about working with integers on a number line, understanding their addition, subtraction, multiplication, and division, as well as how to apply these operations in puzzles and real-world problems.

Recap of Integers and Number Games

The chapter starts with interesting puzzles, like finding two numbers whose sum is 25, and their difference is 11— a great way to practice forming and checking pairs of integers. To solve such problems, you list possible pairs and check which satisfy both conditions. For example, if the sum is 25 and the difference is 11, the numbers are 18 and 7. Similarly, changing the order gives a difference of -11.

- Sum and difference clues can be used for both positive and negative numbers.
- Playing with pairs strengthens number sense and logical reasoning.

Addition and Subtraction Using Real-Life Models

The “carrom coin” activity models integer movements on a number line. A move to the right is positive; a move to the left is negative. The final location is found by adding the effect of each move. For example, a +5 move followed by a -7 move gives a position of -2. The model highlights both direction and magnitude in calculations.

- Removing or adding tokens can help visualize addition and subtraction of integers.
- Subtraction is the same as adding the additive inverse. For example, $7 - 18 = 7 + (-18) = -11$.

Green tokens are used to represent +1, and red tokens represent -1. Together, a green and red token make a zero pair, emphasizing that $(+1) + (-1) = 0$. This helps explain how to

handle large subtractions when there are not enough positives; you add zero pairs to make subtraction possible.

Multiplication of Integers Explained

You learn to multiply integers using the token model, where putting tokens in an empty bag several times captures the multiplication process. Placing 2 positives 4 times is $4 \times 2 = 8$; placing 2 negatives 4 times is $4 \times (-2) = -8$. When the multiplier is negative, you remove tokens. For both negative factors, removing negatives repeatedly leaves you with positives: $(-4) \times (-2) = 8$.

- Multiplying two negatives gives a positive.
- Multiplying a positive with a negative gives a negative.

Patterns are explored with multiplication tables, helping students see trends like flipping signs with multiplication by -1 and the commutative property ($a \times b = b \times a$). For instance, both $3 \times (-4)$ and $(-4) \times 3$ equal -12 . Testing more examples shows that order does not change the product.

Rules for Multiplying and Dividing Integers

Ancient mathematician Brahmagupta described that multiplying or dividing two numbers of the same sign (both positive or both negative) yields a positive result. Mixing one positive and one negative produces a negative result. These simple sign rules are key when working with larger expressions.

- $a \times b = b \times a$ (commutative property)
- $a \times (b \times c) = (a \times b) \times c$ (associative property)
- $a \times (b + c) = a \times b + a \times c$ (distributive property)
- $-1 \times a = -a$ for any integer a

Division of Integers and Related Patterns

Division problems are approached by asking what number multiplied by the divisor gives the dividend. The sign rules for division are the same as for multiplication:

- Quotient is positive when dividend and divisor have same sign.
- Quotient is negative when signs are different.

For example, $(-100) \div 25 = -4$, and $(-100) \div (-4) = 25$. These patterns hold for all integer pairs (except for division by zero).

Applying Integer Operations in Real-life Scenarios

Examples include exam marking schemes (positive for a correct answer, negative for a wrong answer), moving elevators (up as positive, down as negative), changes in temperature over time, and profits and losses in business. Expressions often blend several operations, such as $30 \times 5 + 20 \times (-2) = 110$, or describe changes over time, such as starting from 32°C and dropping 5°C every hour.

Properties and Patterns in Integer Expressions

Multiplication and addition of integers follow special rules:

- Grouping numbers in any way does not affect the result (associative law).
- Order of multiplication does not change the product (commutative law).
- Sign of product with several integers: Even number of negative signs results in a positive product; odd number results in a negative product.
- Distributive property connects multiplication and addition: $a \times (b + c) = a \times b + a \times c$.

Exploring with Activities and Puzzles

Various table-based puzzles and “magic grids” are given to encourage logical thinking about signs, patterns, and calculation methods. Practice with these activities helps in recognizing rules and shortcuts.

- Create possible pairs for given sums and differences.
- Work out calculations using token model (visual aids like colored tokens for plus and minus).
- Test properties (commutative, associative, distributive) using different number sets.

Summary of Key Points

- **Same sign:** Positive result ($- \times -$ or $+ \times +$).
- **Different signs:** Negative result ($+ \times -$ or $- \times +$).
- Multiplication and division rules for signs are consistent.
- **Properties:** commutative, associative, and distributive all hold for integers.
- Real-life contexts like games, shopping, weather, and exams use these concepts directly.

Traditional Games and Logical Thinking

Traditional board games like Terhüchü help develop logical reasoning and strategic thinking, just as pattern puzzles and arithmetic games in maths do. Learning these integer concepts sets the groundwork for later algebra and daily mathematical decisions.